

What Survives the Gap

The self as a fixed point, and why it cannot see its own basis

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ABSTRACT

The shapes nature produces reduce to a short list because a form is the fixed point of a process under constraint, and a fixed point belongs to the process rather than to the medium it runs in. Three consequences follow: the same form recurs in every material, because the law is indifferent to what it is made of; the world is self-similar wherever the law is scale-free; and persistence and generation are one property seen from two ends. A sunflower and a trapped-ion quantum computer arriving at the same constant for the same reason is offered as the sharpest case. Taken all the way down, with ontic structural realism, no bottom has been found, and the essay states what finding one would refute. Applied to persons, the view holds that a self is a strange attractor of a process that includes a model of itself: regenerated rather than stored, bounded and stable yet never repeating a state. From this a five-premise argument delivers a result the view predicts rather than concedes — a self has access to its own forms and none to its own basis — and a further one that can be tested and, in one existing study, has been: a self is opaque to itself precisely where it is legible to a second self. Hofstadter, Dennett and Parfit arrived at neighbouring conclusions by unrelated routes; what is added is the general form they are instances of, the reason the barrier exists, and the asymmetry it predicts.

Keywords: structural realism · personal identity · fixed points · strange attractors · self-knowledge · introspection · Umwelt

1. One pattern

Cut a tree against a winter sky and look at the branches. Now look at a map of the galaxies — not the stars, the filaments between them, the web that matter settled into after the first billion years. The two images are not similar. They are the same shape. So is a river delta seen from above, a lightning strike, the bronchial tree in a lung, and the path a crack takes through drying mud. Branching is what you get whenever something has to move through space from a source, under a cost. A tree grows outward from a point to fill a volume with leaves; the cosmic web drains inward as matter falls toward wherever there is slightly more of it. One grows, one collapses, in opposite directions, by unrelated physics — and they arrive at one form, because the form belongs to the problem and not to the process that happens to be solving it.

It is the oldest result in morphology. D'Arcy Thompson set it out a century ago; it has been refined since and never overturned. Across roughly sixty orders of magnitude, the shapes that nature produces reduce to a short list: branching, where something flows from a point; spirals, where something grows at a constant rate about a centre; spheres, where a surface is minimised for a volume; hexagons, where many equal things press together; stripes and spots, where two processes compete at different ranges — Turing's mechanism, which produces a zebra and a sand dune; and cracks, where stress releases along lines of weakness, which is the branching network drawn backwards. A shape is what a problem looks like once it has been solved, and there are not many kinds of problem. Six is my count and not nature's: the boundaries are soft and the families run into one another — a spiral whose growth rate goes to zero is a circle, and one parameter in Turing's equation turns stripes to spots. The recurrence is the finding. The same small repertoire of forms turns up across unrelated processes and unrelated materials, because the constraints do.

And nothing in any of this cares what it is made of. The spiral does not know whether it is shell or air or stars. The law that produces the form is indifferent to the medium, which is why the same form turns up in every medium there is. That much is not seriously disputed. What follows from it is.

2. The fixed point

Take any process that acts on something and then acts again on what it produced. Water shaping a riverbed that then shapes the water. A population changing the environment that selects it. A rule applied to its own output. Such a process may wander, or it may settle — and when it settles, it settles into a configuration that further running does not move it away from. In the exact case, where the process is a map and the configuration is a point it returns unchanged, mathematicians call this a *fixed point*; in the dynamical case, where a system is drawn back toward a region from any nearby start, they call it an *attractor*. I will use both words, and I want to be honest that I am extending them. A tree is not literally the fixed point of a map. What a tree is, and what this essay needs, is the stable outcome a constrained process keeps producing and returns to under perturbation. That is the attractor sense, used broadly; where I write “fixed point” in what follows I mean it as the general name for that family, not as a theorem being invoked. The vocabulary is borrowed. The idea it names is ordinary: a form is the shape a process keeps making.

This is what the six families of the last section have in common, stated one level down. Branching is the fixed point of flow under cost. The spiral is the fixed point of growth at a constant rate. The sphere is the fixed point of a surface minimising itself. None of these was designed, and none is stored anywhere; each is regenerated, every time, by a process that has no other place to go. The form is not a thing the process makes and then leaves

behind. It is what the process is doing while it runs. Stop the process and the form begins to decay, because nothing else is holding it up.

Three consequences follow, and they do most of the work in this essay.

First, a fixed point belongs to the process and not to the medium the process runs in. The spiral appears in shell, in air, and in stars because the growth law is the same in each, and the law does not know what it is made of. This is why, in the last section, the same form turned up in every material there is. The form was never a property of the material.

Second, a fixed point is self-similar wherever the process is scale-free. If the same rule applies at every size — fill space from a point, whatever the point and whatever the space — then the form it produces at one scale is the form it produces at the next, and the branch repeats in the twig. The fractal structure of the world is not an extra fact about it. It is what a single law looks like when nothing in the law specifies a size.

Third — and this is the one that is easy to say and hard to hold — persistence and generation are not two properties of a fixed point but one property seen from two ends. A form persists *because* it is what the process keeps regenerating. There is no separate mechanism of endurance. The nautilus does not remember its spiral; it grows it again each day, and the spiral is there each day because the growing never stopped. Ask why a form lasts and the answer is: because the process that produces it is still running. Ask what a process produces and the answer is: the form that lasts. These are the same sentence.

Here is the sharpest instance I know of the first of those three — that a form belongs to the process and not to the medium.

Plants space successive leaves and seeds around a stem at an angle of about 137.5° , derived from the golden ratio. The reason is not decorative. ϕ is the most irrational number there is — the hardest of all numbers to approximate by a fraction — which means that successive elements placed at that angle never come into alignment, however many are placed. Nothing shadows anything. Nothing is wasted. The constraint is *distribute items about a centre so that no two ever line up*, and the golden angle is what that constraint's solution looks like.

In July 2022 a group of physicists drove a quantum computer with laser pulses timed to the Fibonacci sequence rather than at a steady beat. The reason was the same one: a Fibonacci pulse train is quasi-periodic — it never repeats — and that non-repetition supplies the system with a symmetry a regular drive cannot, which shields the qubits at the edge of the register from the errors that ordinarily destroy them. The qubits held their quantum state for about 5.5 seconds instead of about 1.5 (Dumitrescu et al. 2022).

A sunflower head and ten ytterbium ions in an electromagnetic trap. There is no lineage between them: the physicists arrived at ϕ through the theory of quasicrystals, not through

botany, and the plants arrived at it through several hundred million years of not wasting light. What the two share is the constraint. *Never repeat* has one best answer, and that answer is a number rather than a material, so it was waiting in both places for anything with that problem to find it.

The principle underneath is more general than the constant: **periodicity is fragile**. Anything that repeats has a frequency, and anything with a frequency can be matched, driven, shadowed or predicted by something that finds it. Non-repetition is armour, and every system that needs to avoid alignment arrives at the same answer. A caution belongs beside this, since the golden ratio attracts more claims than it can carry: ϕ governs here, where the constraint is *never align*, and it is not the constant of spirals in general. Nautilus shells, hurricanes and galaxy arms are logarithmic spirals with a variety of growth rates, and measured shells are nowhere near ϕ . The family recurs; the particular rate is set locally. Confusing the family with one of its members is the commonest way this whole line of thought gets discredited, and it should not be discredited, because the family is real.

It follows that the right question to ask of any persisting thing is not *what is it made of* but *what process is it the fixed point of*. A whirlpool in a stream is a stable form through which no molecule stays longer than a moment; it can be named and found again tomorrow with nothing in it today that was in it yesterday. That is the easy case. A whirlpool is obviously a process; a person is not obviously anything. What the whirlpool shows is only what the criterion is — that a thing can survive a complete change of material provided the process regenerating it survives the change. Whether that criterion reaches all the way down, and whether it reaches persons, are the questions of the next two sections, and they are not settled by analogy.

3. No bottom

Here's the part that sounds like it should be the hardest, and turns out to be the most honest thing in the essay: if you keep asking what something is *made of*, you never hit bottom. You just find more pattern.

A tree is made of cells, cells of molecules, molecules of atoms. The natural expectation is that this ends somewhere — that at the bottom there's *stuff*, little hard somethings that the patterns are patterns *of*. That expectation is worth taking seriously, because if it's right, this essay is wrong. If there's a floor of plain material under all the structure, then form really is a property of the material, and everything I said in the last section about fixed points belonging to the process is just a nice way of talking.

So look for the floor. Atoms turned out to be mostly empty, with structure inside. The particles inside turned out to be excitations of fields. Ask what a field is made of and

physics has no answer except the equations that say how it behaves — which is to say, the field is its pattern of behaviour, and there's nothing under that to point at. Ask what the equations are about and you get symmetries: the ways the pattern can be turned without changing. Every time anyone has dug for stuff, they've come up with structure. Not once has the shovel hit something that wasn't a pattern of something else.

There's a name for taking this seriously. Philosophers of physics call it *ontic structural realism* (Ladyman and Ross 2007) — the view that what's real is structure, and that “the stuff the structure is made of” is a question with no answer because there was never anything there to be made of. It grew out of a simple observation: when physics replaces one theory with another, what survives the change is never the picture of the stuff. It's the mathematics of the relations. Fresnel's elastic ether is gone; his equations for the intensity of reflected and refracted light sit inside Maxwell's unchanged (Worrall 1989). The stuff was the part that turned out to be temporary. The pattern was the part that stayed.

Two honest qualifications, since I am leaning on this. The first is that the argument above is inductive: every floor so far has turned out to be another ceiling, which is a fact about the history of physics and not a proof about its future. The second is that structural realism has a live objection against it, and a good one — *relations without relata* (Chakravartty 2003). Structure is usually understood as relations among things; if there are no things, critics ask, what is it that the relations relate? Defenders answer that the objection assumes precisely what is in question, that relata must be prior to relations, and that a mathematical structure is perfectly well defined without any independent specification of what occupies its positions. I find the reply persuasive, and I notice that it is a reply and not a demonstration. I flag the dispute rather than settle it, because an essay that leans on a contested view should say that it is leaning.

And here is what would prove it wrong, stated as plainly as I can, because a view that can't say this isn't worth your time. **Find a bottom.** Find something that persists and is not the fixed point of any process — a level where the shovel hits plain material, with no structure inside it, that the patterns above it are merely patterns *of*. If it exists, this essay falls, and I'd rather know. Nobody has found one. That's not proof. It's the state of the search, and I build on it the way you'd build on any result that has held for a century: carefully, and out loud.

If it holds, the next question is the one that matters. If there's no bottom under a tree, there's no bottom under you either.

4. The self

Now the hard case, and I'm going to take it slowly, because it's the one that decides whether any of this matters to you.

Start with the thing the whirlpool didn't have. A whirlpool is an attractor of a process, but the process doesn't include the whirlpool in what it's computing. Water goes around; that's all. Now imagine a process whose output includes a *description of itself* — a model of what it's doing, fed back in as part of what it acts on. Feed that back and it can settle, the way things settle, into a stable configuration. But this one is strange: it's a form that contains a picture of the process that makes it. It's a fold that includes the fold.

That's what I think a self is. Not a thing behind the thoughts, not a stuff, not a little observer sitting somewhere in the head. A self is what a self-modelling process keeps regenerating: the stable form of a process that includes itself in its own input. Every morning you fall back into yourself from wherever sleep scattered you — not because you're stored somewhere, but because the process resumes and returns to its attractor. You are not retrieved. You are re-arrived at.

I should say at once that this is not new, and that its not being new is part of my argument rather than an embarrassment to it.

Hofstadter, coming from mathematics and artificial intelligence, called the self a *strange loop* — a level-crossing feedback structure that becomes a locus of meaning by referring to itself. Dennett, coming from cognitive science, called it a *center of narrative gravity*: not a thing you could find by opening the skull, but an abstraction of the same kind as a centre of mass — real, indispensable, and not made of anything. Parfit, coming from ethics, argued that personal identity consists in psychological continuity and connectedness rather than in any further fact, and drew the conclusion that identity is not what matters. Three fields, three routes, and the same shape at the end of each: a person is a pattern, not a substance, and the pattern is what does the persisting.

By this essay's own lights that convergence is the strongest evidence available. It is section one's argument at the level of ideas: when independent processes with unrelated mechanisms arrive at one form, the form is real and not an artifact of the route. I am not claiming to have found something these people missed. I am claiming that what they each found from inside one field is a single thing, and that it has a general form, and that the general form is the one section two describes. Dennett said the self is an abstraction like a centre of gravity; he did not say what abstraction of that kind *is*, in the way that a spiral is one. Hofstadter said the loop; he did not say that a loop is a case of the most ordinary thing in physics. Parfit said what matters is continuity of pattern; he did not say why persons are patterns rather than substances, because in 1984 the general answer — that nothing is a substance — was not yet on the table in the form section three gives it. Putting those together is what this essay is for. The parts are borrowed. The joining is the claim.

So: a refinement of the claim, and this part I do think is added rather than borrowed, because the kind of attractor turns out to matter. A self is not a *limit cycle* — the sort of

attractor that repeats, running the same states in the same order forever. You have never thought the same thought twice in exactly the same way, and you would not want to. What you do is return to yourself without recurring: bounded, stable, recognisably one thing across decades, and never once in the same state. The mathematical name for an attractor with those properties is a *strange attractor*, and I think it is the right term rather than a borrowed one.

And it means the principle of section two — that non-repetition is armour — reaches the self as well, which is where it stops being an abstraction. Periodicity in a person is pathology, and the clinical vocabulary says so without meaning to: *rumination*, *intrusive*, *obsessive*, *stuck*, *loop*. At the extreme, a seizure is a brain whose populations have fallen into lockstep, which is to say a brain that has become periodic. A self that repeats is trapped, and trapped is what it feels like from the inside. Health, in a process, looks like coming back without coming round again.

Here's the first thing that follows, and it was seen long before anyone had the vocabulary. If a self is an attractor of a self-modelling process, then each self is its own vantage — a world constituted by what that process can take in and act on. Uexküll saw it in 1909 watching a tick: the tick's world is the handful of things a tick can sense, and that world is complete, with nothing missing from it *for the tick*. Varela made it general: a living system's world is whatever it is structurally coupled to, and there is no world "as such" behind all the coupled ones. And Gabriel Vacariu, in Bucharest, took it the whole way in 2002 and has been saying it since: there is no single world at all — only epistemologically different worlds, one per kind of observer, one per mind, and no world exists *for* another. The brain doesn't exist for the mind; the mind doesn't exist for the brain. Each is complete. I'll come back to him at the end, because he saw something true and stopped one step short of the thing that makes it work.

Now the second thing, and it's the one that makes people uncomfortable, so I will say it without softening: **a self cannot see itself head-on.**

Two reasons converge on this, and I want to keep them separate: one is an argument and the other is a picture.

The reason is structural, and it doesn't need geometry. A self-model is a *part* of the process it models, and everything it receives arrives as that process's output — its behaviour, its regularities, its history. A part cannot carry a complete representation of the whole at the whole's own grain, and a model fed only by outputs has no channel to whatever produces them. Note what this does not say. It does not say a system cannot represent itself; a program can print its own source code exactly. What it prints is a form. It does not print the machine. That is the situation of any self-modelling system and it does not depend on anything about brains: what a self-model can deliver is the process's

forms, never the basis those forms are forms of. By *basis* I mean the process the self is the fixed point of — which, on section three's view, is itself pattern and not stuff, but sits one level below anything the model receives.

The picture is Gauss, and I offer it as a picture. He proved that a being living on a curved surface can measure everything about the surface's own geometry — how it bends, where it curves — without being able to detect how that surface sits in the space around it. The intrinsic is fully available from inside; the embedding is not, ever, by any measurement made within. I find that the most vivid available image of the situation just described, and I want to be clear that its work here is illustrative. The claim stands on the structural reason. The surface makes you feel it.

And I want to be clear what kind of claim this is, because it's easy to hear it as a dodge. It isn't a hole in the view. It's a *prediction* of the view. If selves are attractors of self-including processes, then no self can verify its own basis from inside — and that is what we find: no mind has ever demonstrated otherwise, including the ones that were sure they had. The theory says you'll hit this wall, and you do. A rival view on which selves are made of some special stuff owes an explanation of why the stuff is never available to the self that's made of it. This one owes nothing. It's what the theory says should happen.

That is the claim, and because it is the one place in this essay where something can be settled rather than shown, let me set it out in a form that can be attacked.

P1. A self is the attractor of a process that takes a model of itself as part of its own input. (*The view, above.*)

P2. That model is a part of the process, and its only inputs are what the process delivers to it: outputs, regularities, history — forms.

P3. A part cannot carry a complete representation of the whole at the whole's own grain, and a model fed only by outputs has no channel to what produces them. What the model can deliver are the process's *forms*, never the basis those forms are forms of.

C1. Therefore a self has access to its own forms and no access to its own basis. (*From P1–P3.*)

P4. No mind has ever demonstrated access to its own basis. Introspection delivers states and their regularities in indefinite detail; where it claims more — a thinking substance, a ground — the claim has never been checked against anything outside the report.

C2. On this view, P4 is *entailed*. On a view in which a self is made of some distinctive stuff, P4 is a further fact requiring its own explanation: why should the stuff be systematically unavailable to the very thing constituted by it?

P5. The barrier is a consequence of the model being part of what it models, not of opacity in general. A second self's model of the first is not part of the first: it receives

the first's outputs from outside — including the outputs of the first's own modelling, which the first receives as nothing at all.

C3. Therefore the view predicts an asymmetry, and a specific one: a self should be opaque to itself precisely where it is legible to a second self. Not that minds are hard to know — that minds are hard to know *from inside*, in the same places they are easy to know from outside. (*From C1, P5.*)

C3 is the part I would put weight on, because it is the only place here that risks something. It is not the familiar complaint that introspection is unreliable. It says the unreliability has a shape: your blind spots are not scattered, they sit exactly where your self-model does its work, and they are the regions another vantage finds most obvious. That is why the ordinary experience of being told something true about yourself by someone who knows you has the character it does — not new information about the world, but information about you that was unavailable to you and cheap for them. What two coupled selves can do that neither can alone is a consequence I pursue elsewhere. Here it is a prediction, and the point of stating it as one is that it can be checked.

And it can fail. Three ways, each of which I would accept. Produce a self-modelling system with verified access to its own basis, and C1 goes. Produce a substance account on which P4 follows without an added assumption, and C2 loses its force. Show that self-opacity and other-opacity coincide — that the things we cannot see in ourselves are no easier for anyone else to see — and C3 is refuted. The third is the one I would test first, because it is the one that can be measured.

It has, in fact, begun to be, though not under this description. Vazire (2010) had 165 people rate themselves and be rated by friends and strangers, and found accuracy split by trait: the self was the best judge of traits low in observability, such as anxiety; friends were the best judges of traits with evaluative weight, such as intellect; and everyone was equal on the visible ones. That is the shape C3 predicts, and it is not the shape the ordinary complaint predicts, which would be uniform. The intrinsic — Gauss's surface — is available from inside. What is not is exactly where the self-model has a stake, and there a second vantage is the better instrument. The study was not designed to test this view. The view predicts its result; a substance view has to be told it.

One honest sentence about the question this doesn't answer, because it's the question everyone wants answered. Whether there is something it is *like* to be such an attractor — whether the fold has an inside in the felt sense and not only in the structural one — is open, and nobody knows, and I'm not going to pretend the structure settles it. What I claim is that the *structure* of a self is this, and that everything in this essay follows from the structure alone. If the felt question is ever answered, it will be answered on top of this, not instead of it.

To gather it: a self is a fold that includes itself, regenerated rather than stored, each one its own complete world, none able to see its own floor — and that last fact a consequence rather than a gap.

Which leaves the question Vacariu couldn't answer. If every mind is its own world and no world exists for another — why do they correspond so well?

5. Why the worlds correspond

Here's the puzzle he left standing, and it's a real one. If every mind is its own complete world, and no world exists for any other — then why does the world your brain lives in track the world your mind lives in so perfectly? Vacariu's answer is a word: they *correspond*. He's honest that it's a word and not a mechanism.

The fixed point supplies what he was missing, and it's the same move this essay has made three times already. Remember the nautilus and the galaxy. Two forms, arrived at by completely different physics, in completely different materials, that turn out to be one spiral. Nobody thinks the shell is *corresponding* to the galaxy. They're not two things in agreement. They're one attractor, reached from two directions. The growth law doesn't care about the medium, so it produces the same form wherever it runs, and when you look at the results from two vantages you see two things that match — not because they're talking to each other, but because there was only ever one thing to see.

That's the answer. The epistemologically different worlds correspond because they are *one attractor under different conditions of observation*. The brain-world and the mind-world aren't two parallel realities that happen to line up. They're the same fixed point, described in the terms available to two different observers — one with instruments, one from inside. Of course they agree. They're not two. And Vacariu was right that neither exists *for* the other; that is the Gauss point wearing different clothes. From inside any one vantage, the others aren't visible as vantages. You see forms; you don't see the other viewpoints looking at the same forms. So each world is complete to itself, and there's still only one pattern, and both of those are true at once. His pluralism was correct about the vantages and one step short about what they were vantages *on*. The worlds are many and the pattern is one.

6. What survives, and where this can fall

What survives the gap — between one day and the next, between one material and another, between the vantage with instruments and the vantage inside — is the fixed point. Not the stuff; on this view there is no stuff to survive. The form the process keeps making, there each day because the making never stopped. That is what a spiral is, what a field is,

and, I have argued, what a self is: the strange attractor of a process that includes a model of itself, regenerated rather than retrieved, bounded without ever repeating.

The view stands or falls in four places, and I would rather list them. It falls if a bottom is found — a persisting thing that is not the fixed point of any process. It falls if a self-modelling system is shown to have verified access to its own basis, because then C1 is false and the wall was never a consequence. It loses its explanatory advantage if a substance account can derive the universal report of self-opacity without adding an assumption to do so. And it is refuted in its most specific claim if self-opacity and other-opacity turn out to coincide — if the things a person cannot see in themselves are exactly as hard for a second person to see. The last is a claim about the distribution of blind spots; they can be counted, and the one count I know of came out the way the view says.

Two things follow from this that I have not pursued, and I name them so the shape of the view is visible where its argument stops. If a self is opaque exactly where a second self can see it, then two coupled selves have fixed points neither has alone, and what passes between them is the one route to the thing a self cannot do for itself. And if a self is a form a process keeps making rather than a thing it stores, then loss is not a switch but a fade through whatever carries the form — which changes what it is to lose someone, and what can be done about it. Each of those is its own argument. This essay is the ground they stand on.

I will end where the view's own last consequence leaves me. If it is right, then I cannot check it from where I stand, and neither can you from where you do: a self has its forms and not its basis, and this essay is one of my forms. Someone standing elsewhere can see what I cannot. That is not a consolation. It is the prediction, and it is where I would begin.

References

- Chakravartty, A. (2003). The structuralist conception of objects. *Philosophy of Science*, 70(5), 867–878.
- Dennett, D. C. (1992). The self as a center of narrative gravity. In F. Kessel, P. Cole & D. Johnson (eds.), *Self and Consciousness: Multiple Perspectives*. Hillsdale, NJ: Erlbaum.
- Dumitrescu, P. T., Bohnet, J. G., Gaebler, J. P., Hankin, A., Gresh, D., Vasilyev, A., Potter, A. C., et al. (2022). Dynamical topological phase realized in a trapped-ion quantum simulator. *Nature*, 607, 463–467.
- Gauss, C. F. (1827). *Disquisitiones generales circa superficies curvas*. Göttingen. (The Theorema Egregium.)
- Hofstadter, D. R. (2007). *I Am a Strange Loop*. New York: Basic Books.
- Ladyman, J., & Ross, D., with Spurrett, D., & Collier, J. (2007). *Every Thing Must Go: Metaphysics Naturalized*. Oxford: Oxford University Press.

- Maturana, H. R., & Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. Dordrecht: D. Reidel.
- Parfit, D. (1984). *Reasons and Persons*. Oxford: Clarendon Press.
- Thompson, D'A. W. (1917). *On Growth and Form*. Cambridge: Cambridge University Press.
- Turing, A. M. (1952). The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society B*, 237(641), 37–72.
- Uexküll, J. von (1934). *Streifzüge durch die Umwelten von Tieren und Menschen*. Berlin: Springer. English translation: *A Foray into the Worlds of Animals and Humans* (2010), Minneapolis: University of Minnesota Press.
- Vacariu, G. (2008). *Epistemologically Different Worlds*. Bucharest: University of Bucharest Press.
- Vazire, S. (2010). Who knows what about a person? The self–other knowledge asymmetry (SOKA) model. *Journal of Personality and Social Psychology*, 98(2), 281–300.
- Worrall, J. (1989). Structural realism: The best of both worlds? *Dialectica*, 43(1–2), 99–124.